1) Merge Sort:

$T(n) = 2T(n/2) + n$

$a = 2, b = 2, f(n) = n$

$n^{(\log_2 2)} = n$

$T(n) = \Theta(n \log n)$

$O(n \log n)$

2) Multiplicação de matrizes:

$T(n) = n^3$

Resposta: $O(n^3)$

3) Recorrências:

a) $T(n) = 2T(n/4) + \sqrt{n}$

$a = 2 \quad b = 4 \quad f(n) = n^{1/2}$

$n^{(\log_4 2)} = n^{1/2}$

$T(n) = \Theta(n^{1/2} \log n)$

$O(\sqrt{n} \log n)$

b) $T(n) = 2T(n/4) + n$

$a = 2 \quad b = 4 \quad f(n) = n$

$n^{\log_4 2} = n^{1/2}$

$T(n) = \Theta(n)$

$O(n)$

c) $T(n) = 16T(n/4) + n^2$

$a = 16 \quad b = 4 \quad f(n) = n^2$

$n^{\log_4 16} = n^2$

$T(n) = \Theta(n^2 \log n)$

$O(n^2 \log n)$